

Report
Of

**CLOUD COMPUTING INTERNSHIP:
DEVELOPMENT OF A CLOUD-BASED FILE STORAGE SYSTEM**

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

By

Roll Number of Student: 25SCS1003003052

Name of Student: RISHAV ANAND

Batch: 2025-29

CANDIDATE'S DECLARATION

I, RISHAV ANAND do hereby declare that the internship report titled **“Cloud Computing Internship: Development of a Cloud-Based File Storage System”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: RISHAV ANAND

Roll No.: 25SCS1003003052

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Student Name: RISHAV ANAND

Roll No.: 25SCS1003003052

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CHAPTER 3

INTERNSHIP COMPLETION CERTIFICATE



Internship Certificate issued to Rishav Anand by Codec Technologies Pvt. Ltd.

CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

The internship report titled “**Cloud Computing Internship: Development of a Cloud-Based File Storage System**” documents the practical learning and professional exposure gained during the internship at **Codec Technologies Pvt. Ltd.**. The internship was completed from **29 July 2026 to 29 August 2026** in the role of **Cloud Computing Intern**, carried out in Pan India, Hybrid mode and reporting to the assigned Project Head(s).

The work focused on gaining industry-level knowledge and hands-on skills in cloud computing through an assigned project. As part of the internship's practical component, the intern was assigned the design and planning of a **Cloud-Based File Storage System**, a secure file storage application similar in concept to Google Drive.

This project description presents the internship from an academic and technical perspective. It explains the organizational context, the problem addressed, objectives, scope, tools, system architecture, methodology and expected outcomes. The report is based on the responsibilities and skills documented in the official internship offer letter and completion certificate.

Project Theme: Design and planning of a secure, scalable cloud-based file storage and sharing system using cloud storage services and modern authentication practices.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. is the organization where the internship was completed. The official internship offer letter identifies Codec Technologies as a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

The internship was conducted under the **National Internship Portal**, an AICTE and ICAC approved program, and Codec Technologies is recognized as a Google for Education Partner and an ISO 9001:2015 certified organization.

The organization's office is located at Chandivali IT hub, Mumbai, and the internship program was issued and coordinated by Dr. Anurag Shrivastava (Talent Acquisition Manager / Program Manager). The internship was structured with weekly performance reviews conducted by the project supervisor, with exceptional interns qualifying for a scholarship award.

4.3 Problem Statement

Individuals and organizations generate increasing volumes of digital files that need to be stored, accessed and shared securely from anywhere. Local storage is limited, prone to data loss, and difficult to share across devices and teams, while ad-hoc sharing methods often lack proper access control and version history.

The project therefore addressed the following problem: **how to design a secure, cloud-based file storage system, similar to Google Drive, that allows users to upload, store, retrieve and share files reliably while ensuring that only authenticated and authorized users can access them.**

The problem was approached from a cloud-computing design perspective, focusing on selecting an appropriate cloud storage backend, defining a secure authentication mechanism, and planning features such as file versioning and controlled sharing, in line with the guidelines provided for the assigned project.

4.4 Project Objectives

1. **Build a secure cloud storage application:** Design an application similar in concept to Google Drive that lets users store and access files securely.
2. **Evaluate cloud storage backends:** Study and compare AWS S3, Azure Blob Storage and Google Cloud Storage to select a suitable storage backend.
3. **Implement authentication:** Plan an OAuth 2.0-based authentication flow so only verified users can access their files.
4. **Support file versioning:** Allow multiple versions of the same file to be retained and restored when needed.
5. **Enable controlled file sharing:** Provide sharing options so users can share files or folders with specific permissions.
6. **Develop professional and technical skills:** Strengthen cloud computing, problem-solving and documentation skills gained during the internship.
7. **Understand industry cloud practices:** Gain industry-level knowledge of how cloud storage systems are designed and secured in practice.

4.5 Scope of the Project

Area	Scope Covered
Cloud Storage Backend	Evaluating and selecting among AWS S3, Azure Blob Storage and Google Cloud Storage.
Authentication & Security	Planning OAuth 2.0-based authentication for secure user access.
File Versioning	Designing support for maintaining and restoring previous file versions.

File Sharing	Designing controlled sharing options for files and folders.
Documentation	Preparing project documentation covering design decisions and workflow.
Professional Development	Building cloud computing, problem-solving and communication skills.

The scope does not claim ownership of a fully deployed production system unless specifically documented elsewhere; the internship evidence primarily supports a design, planning and cloud-fundamentals learning role.

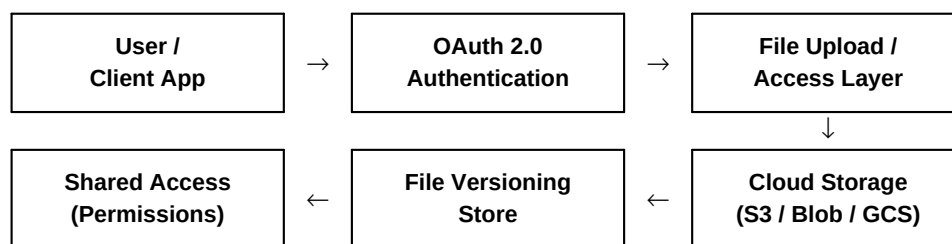
4.6 Technologies and Tools Used

Tool / Technology	Use in the Project Context
AWS S3 / Azure Blob / Google Cloud Storage	Candidate cloud storage backends evaluated for storing files.
OAuth 2.0	Authentication protocol planned for secure user login and access control.
File Versioning Mechanism	Approach for retaining and restoring earlier versions of stored files.
Sharing & Access Control	Permission-based mechanism for sharing files and folders with other users.
Documentation Tools	Used for maintaining project documentation and reports.

In addition to tools, the internship required process skills such as cloud fundamentals, security awareness, problem-solving and documentation. These skills align with the Cloud Computing Intern role recognized in the completion certificate.

4.7 System Architecture

The architecture below represents a **conceptual system architecture** for the Cloud-Based File Storage System designed during the internship. It shows how a user's request moves from authentication through storage and sharing.



Conceptual architecture designed for the Cloud-Based File Storage System project.

Workflow interpretation: A user authenticates via OAuth 2.0, after which requests are routed through the file upload/access layer to the selected cloud storage backend. Each update is tracked by the versioning store, and access to files is governed by the sharing and permissions layer, allowing controlled collaboration similar to Google Drive.

This architecture is a conceptual design prepared for the assigned project; it is not presented as a deployed proprietary system of Codec Technologies.

4.8 Methodology

Step	Activity	Purpose
1	Understand requirements	Study the objective of building a secure, Google Drive-like storage app.
2	Compare storage backends	Evaluate AWS S3, Azure Blob Storage and Google Cloud Storage.
3	Design authentication	Plan an OAuth 2.0-based login and access-control flow.
4	Plan file versioning	Design how multiple file versions are retained and restored.
5	Plan sharing options	Design permission-based file and folder sharing.
6	Document the design	Record architecture, workflow and design decisions.
7	Review & improve	Use weekly supervisor feedback to refine the design.

The methodology reflects the practical nature of the internship and aligns with the project guidelines: use of a cloud storage backend, OAuth 2.0 authentication, and file versioning and sharing options.

4.9 Expected Outcomes

1. A working understanding of major cloud storage services (AWS S3, Azure Blob Storage, Google Cloud Storage).
2. A secure authentication design using OAuth 2.0 for a cloud application.
3. A design for file versioning that preserves and restores earlier file states.
4. A design for controlled file and folder sharing between users.
5. Improved understanding of cloud computing architecture and industry practices.
6. Stronger problem-solving, documentation and communication skills.
7. Practical, industry-level exposure gained through the Codec Technologies internship program.

The internship offer letter and completion certificate confirm that these areas contributed to the intern's practical skill development, recognizing the role of Cloud Computing Intern completed under the AICTE & ICAC approved National Internship Portal program.

4.10 Certificates of Completion and Other Communication Mail Proof

The official **Internship Certificate** issued by Codec Technologies Pvt. Ltd. is attached as Chapter 3 of this report without alteration. It confirms completion of the 1-month AICTE & ICAC approved internship program as Cloud Computing Intern, conducted from 29/07/2026 to 29/08/2026, under the National Internship Portal.

The following page includes the **Internship Offer Letter** issued by Codec Technologies confirming the designation, duration, mode and reporting structure of the internship. It is reproduced as supporting evidence.

INTERNSHIP OFFER LETTER

Dear Rishav Anand ,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Cloud Computing Intern
- Location & Mode: Pan India, Hybrid
- Duration : 29/07/2026 to 29/08/2026
- Reporting to : Assigned Project Head(s)

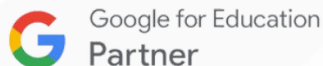
This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 29/07/2026


Dr. Anurag Shrivastava
Talent Acquisition Manager

5. BIBLIOGRAPHY / REFERENCES

1. Codec Technologies Pvt. Ltd. — Internship Certificate issued to Rishav Anand, National Internship Portal (AICTE ID: CORPORATE6759d549ce59e1733940553).
2. Codec Technologies Pvt. Ltd. — Internship Offer Letter issued to Rishav Anand (NCS ID: E19E86-0116588288923), Date of Issue: 29/07/2026.
3. Project brief — “Cloud-Based File Storage System”: objective and guidelines provided for the assigned internship project.

Note: The report content has been prepared around the internship responsibilities and project guidelines documented in the supplied offer letter, certificate and project brief.